

Transformations, Compositions and Inverses Together

Answer Key

Precalculus Advanced function transformations and composition

Quick Answers

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|-------------------|--------------------|-----------------------|
| 1. $x^2 - 4x + 9$ | 5. $x^2 + 8x + 16$ | 9. -1 |
| 2. 7 | 6. x , 6.5 | 10. $\frac{x}{2} - 3$ |
| 3. x , 5 | 7. 8 | 11. -5 |
| 4. $-4x^2 + 2x$ | 8. 0.5 | 12. x , 5 |
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Curriculum

Level: Precalculus Advanced function transformations and composition

HSF-BF.A.1c – *problems 2, 5, 8*

HSF-BF.B.4 – *problems 3, 6, 9, 10, 12*

HSF-IF.A–HSF-IF.C – *problems 1, 4, 7, 11*

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

7. *If they got 2: shifted right instead of left, using x minus 4 inside the root*

11. *If they got -2: applied the vertical stretch to the shift as well, doubling the plus 3*

1. $g(x) = f(x - 2) + 5$ for $f(x) = x^2$.

Step 1. Read the two changes by position: the -2 is inside f , so it is horizontal and moves *opposite* to its sign; the $+5$ is outside, so it is vertical and moves as written.

Step 2. So the graph shifts right 2 and up 5, and the rule is $g(x) = (x - 2)^2 + 5$.

Step 3. Expanding, $(x - 2)^2 + 5 = x^2 - 4x + 4 + 5$, so $g(x) = \boxed{x^2 - 4x + 9}$.

Listen for: “left 2”. Inside the function, the sign lies about the direction — $f(x - 2)$ needs a larger x to reach the same output, so the graph moves right.

2. $(f \circ g)(2)$ for $f(x) = 2x - 3$, $g(x) = x^2 + 1$.

Step 1. Composition is evaluated inside out, so g runs first and its output becomes the input of f .

Step 2. $g(2) = 2^2 + 1 = 5$.

Step 3. Then $f(5) = 2(5) - 3$, so $(f \circ g)(2) = \boxed{7}$.

Common wrong answer: $g(f(2)) = g(1) = 2$ — the two functions were applied in the wrong order. Composition is not commutative.

3. The inverse of $f(x) = 2x - 3$.

Step 1 (a). An inverse undoes f , so write $y = 2x - 3$, swap the roles of the variables to get $x = 2y - 3$, and solve for y .

Step 2 (a). Adding 3: $x + 3 = 2y$. Dividing by 2: $f^{-1}(x) = \frac{x + 3}{2}$.

Step 3 (b). Check by composing: $f(f^{-1}(x)) = 2 \cdot \frac{x + 3}{2} - 3 = (x + 3) - 3 = x$, which is what “undoes” means.

Step 4 (b). Evaluating, $f^{-1}(7) = \frac{7 + 3}{2}$, so $f^{-1}(7) = \boxed{5}$.

Check: $f(5) = 2(5) - 3 = 7$, so the pair $(5, 7)$ on f becomes $(7, 5)$ on f^{-1} — the reflection across $y = x$, seen numerically.

4. $h(x) = -f(2x)$ for $f(x) = x^2 - x$.

Step 1. Two changes again, and they are of different kinds: the $2x$ is inside (horizontal), the leading minus is outside (vertical).

Step 2. Substituting $2x$ for x : $f(2x) = (2x)^2 - (2x) = 4x^2 - 2x$.

Step 3. Negating everything gives $h(x) = \boxed{-4x^2 + 2x}$: a horizontal compression by a factor of one half, followed by a reflection in the x -axis.

Common error: distributing the minus to only the first term. It is outside the whole function, so every term flips.

5. $(g \circ f)(x)$ for $f(x) = x + 4$, $g(x) = x^2$.

Step 1. f runs first, so its whole output $x + 4$ is what gets squared — not x .

Step 2. $(g \circ f)(x) = g(x + 4) = (x + 4)^2$.

Step 3. Expanding, $(g \circ f)(x) = \boxed{x^2 + 8x + 16}$.

What this shows: composing with $x + 4$ on the inside is exactly the horizontal shift rule — the graph of $g \circ f$ is the parabola $y = x^2$ moved left 4. Transformations are compositions with simple linear functions.

6. The inverse of $f(x) = \frac{2x + 1}{x - 3}$.

Step 1 (a). Same procedure as any inverse: $y = \frac{2x + 1}{x - 3}$, swap to get $x = \frac{2y + 1}{y - 3}$, then solve for y — but now the unknown sits in both the numerator and the denominator, so clear the fraction first.

Step 2 (a). Multiplying: $x(y - 3) = 2y + 1$, so $xy - 3x = 2y + 1$.

Step 3 (a). Collect the y terms: $xy - 2y = 3x + 1$, so $y(x - 2) = 3x + 1$ and $f^{-1}(x) = \frac{3x + 1}{x - 2}$.

Step 4 (b). Substituting f^{-1} into f and simplifying returns x , which confirms the algebra.

Evaluating, $f^{-1}(4) = \frac{13}{2}$, so $f^{-1}(4) = \boxed{\frac{13}{2}}$.

Note: f excludes 3 from its domain and never outputs 2; the inverse excludes 2 and never outputs 3. Domain and range swap places, just like the coordinates.

7. h from $f(x) = \sqrt{x}$: stretch 3, left 4, down 1.

Step 1 (a). Sort the three changes by whether they touch x or y : a vertical stretch and a vertical shift are outside, and the horizontal shift is inside with the opposite sign.

Step 2 (a). Left 4 means $x + 4$ inside the root; stretch by 3 and down 1 are outside, so $h(x) = 3\sqrt{x + 4} - 1$.

Step 3 (b). $h(5) = 3\sqrt{9} - 1 = 3(3) - 1$, so $h(5) = \boxed{8}$.

Common wrong answer: 2 — from $3\sqrt{5 - 4} - 1$, shifting right instead of left.

8. $(f \circ g)(-1)$ for $f(x) = \frac{1}{x - 2}$, $g(x) = x + 5$.

Step 1. Inside out again: find $g(-1)$ first, then feed it to f .

Step 2. $g(-1) = -1 + 5 = 4$, and $f(4) = \frac{1}{4 - 2} = \frac{1}{2}$, so $(f \circ g)(-1) = \boxed{\frac{1}{2}}$.

Step 3. For the domain: $f \circ g$ is $\frac{1}{x + 3}$, which is undefined when $x + 3 = 0$. The excluded input is $x = -3$.

Teaching note: the excluded value comes from the *inner* output hitting f 's bad input, which is why domains of compositions have to be checked before simplifying, not after.

9. Solving $f^{-1}(x) = 2$ for $f(x) = 3x - 7$.

Step 1. $f^{-1}(x) = 2$ says exactly that $f(2) = x$ — reading the inverse statement backwards is the shortcut, and it avoids finding f^{-1} at all.

Step 2 (shortcut). $f(2) = 3(2) - 7 = -1$, so $x = -1$.

Step 3 (check with the formula). $f^{-1}(x) = \frac{x + 7}{3}$, and setting $\frac{x + 7}{3} = 2$ gives $x + 7 = 6$, so $\boxed{x = -1}$ — the same answer.

What this shows: an inverse equation is the original equation with the roles of input and output swapped. Students who see that stop computing inverses they do not need.

10. $(f \circ g)^{-1}$ against $g^{-1} \circ f^{-1}$, for $f(x) = 2x$ and $g(x) = x + 3$.

Step 1 (a). Compose first: $(f \circ g)(x) = f(x + 3) = 2(x + 3) = 2x + 6$.

Step 2 (a). Inverting that: $y = 2x + 6$ swaps to $x = 2y + 6$, so $y = \frac{x - 6}{2}$.

Step 3 (b). Separately, $f^{-1}(x) = \frac{x}{2}$ and $g^{-1}(x) = x - 3$, so $g^{-1}(f^{-1}(x)) = \boxed{\frac{x}{2} - 3}$.

Step 4. The two agree, since $\frac{x - 6}{2} = \frac{x}{2} - 3$.

Why the order reverses: $f \circ g$ does g first and f second, so undoing it must undo f first and g second — the same reason you take your shoes off before your socks.

11. $h(x) = -2f(x - 1) + 3$ with $f(1) = 4$ and $f(3) = -2$.

Step 1 (a). $h(2)$ needs f at $2 - 1 = 1$, and the table supplies that value — so the only table entry this part uses is $f(1) = 4$.

Step 2 (a). $h(2) = -2f(1) + 3 = -2(4) + 3$, so $h(2) = \boxed{-5}$.

Step 3 (b). In order: shift right 1 (inside, opposite sign); then stretch vertically by 2 and reflect in the x -axis (the factor -2 outside); then shift up 3.

Grading note (manual): part (b) needs the order as much as the list. Applying the up-3 before the -2 would multiply it as well, giving -2 instead of -5 — and that wrong value is the check on whether the order was understood.

Common wrong answer: -2 , from stretching after shifting up.

12. Synthesis: $f(x) = (x - 1)^2$ on $x \geq 1$.

Step 1 (a). On all real numbers f fails the horizontal line test — $f(0) = f(2) = 1$, so two inputs share an output and no rule could undo it. Restricted to $x \geq 1$ the function is strictly increasing, so each output comes from exactly one input and an inverse exists.

Step 2 (b). $y = (x - 1)^2$ swaps to $x = (y - 1)^2$. Taking the square root and keeping the branch with $y \geq 1$ gives $\boxed{f^{-1}(x) = \sqrt{x} + 1}$ — the positive root is chosen because of the restriction, not by convention.

Step 3 (b). Check: $f(f^{-1}(x)) = (\sqrt{x} + 1 - 1)^2 = x$ for $x \geq 0$, so the two really do undo each other.

Step 4 (c). $f(5) = (5 - 1)^2 = 16$ and $f^{-1}(16) = \sqrt{16} + 1 = 4 + 1$, so $f^{-1}(f(5)) = \boxed{5}$.

What that shows: the point $(5, 16)$ on f corresponds to $(16, 5)$ on f^{-1} , and going out and back returns the input. That round trip is the reflection across $y = x$ done in numbers rather than in a picture.

Grading note (manual): part (a) is the graded reasoning here. Either the horizontal line test with a concrete pair of inputs, or “ f is strictly increasing on $x \geq 1$ ”, earns full credit; “because we restricted it” does not.